

CSA15 Cloud Computing and Big Data Analytics

Local IaaS Operations Lab: Snapshot, Clone and Restore Testing

Course Code	CSA15	Course Title	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-03	Submission	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Target Learner	Beginner CSE student	Estimated Duration	2 to 3 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation	Cloud Link	Builds operational recovery thinking before cloud VM backup and image management

Guided Project Pattern

Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom.

Learning Outcomes

- Create a safe baseline snapshot of a configured VM.
- Test snapshot recovery by making a visible temporary change and restoring the previous state.
- Create or plan a full/linked clone according to storage availability and tool support.
- Compare snapshot, clone and VM image/export in operational terms.
- Prepare a complete VM recovery evidence portfolio.

Before You Start

- Completed VM from GP-VM-01 and configuration/storage evidence from GP-VM-02.
- At least 5 GB free storage for clone testing where possible; otherwise prepare a justified clone plan.
- The VM must not contain passwords, keys or personal sensitive data in screenshots.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A snapshot records a point-in-time VM state and is useful before risky changes.
- A clone creates another VM copy for parallel testing or repeated experiments.
- A full clone is independent and consumes more storage. A linked clone depends on the base VM and usually consumes less storage.
- A restore test is important because a snapshot is useful only if it can actually return the VM to an earlier state.

Lab Exercise Coverage

Lab Exercise	How This Guided Project Covers It
Experiment 10	Create a snapshot and test it by loading/restoring the previous version.
Experiment 11	Create a clone of a VM and test it by opening the cloned VM.

Hypervisor Choice

Tool	When to Choose It	Student Evidence
Oracle VirtualBox	Use Snapshots from the VM tools/menu and Clone from Machine options.	Snapshot list, restore test, clone window and cloned VM proof.
VMware Workstation	Use Snapshot Manager / Take Snapshot and VM -> Manage -> Clone where available.	Snapshot manager, revert proof, clone wizard and cloned VM proof.
Safety Rule Do not delete the original VM while testing snapshots or clones. If storage is low, record a clone plan with menu evidence instead of creating a large duplicate.		

Architecture View

Base VM	Baseline Snapshot	Temporary Change	Restore Test	Clone/Plan	Recovery Evidence
CSA15-UbuntuServer-192472346	csa15-clean-baseline-192472346	csa15-after-snapshot.txt file created	Restored baseline state (after marker deleted)	Full Clone (CSA15-Clone-192472346)	Screenshots 01-05 verified (ls logs & VirtualBox snapshot manager)

The recovery workflow begins with a stable VM, captures a baseline, performs a controlled change, restores the previous version and then prepares a duplicate environment or clone plan.

Guided Build

Stage 1 - Prepare the Baseline VM

Learn: Snapshot and clone testing must begin from a known working VM state.

- Open the VM in VirtualBox or VMware.
- Boot the guest OS and confirm that it reaches desktop, shell or login prompt.
- Create a visible marker file inside the guest OS if possible, such as csa15-before-snapshot.txt.
- Shut down or pause the VM according to the hypervisor recommendation.

Checkpoint

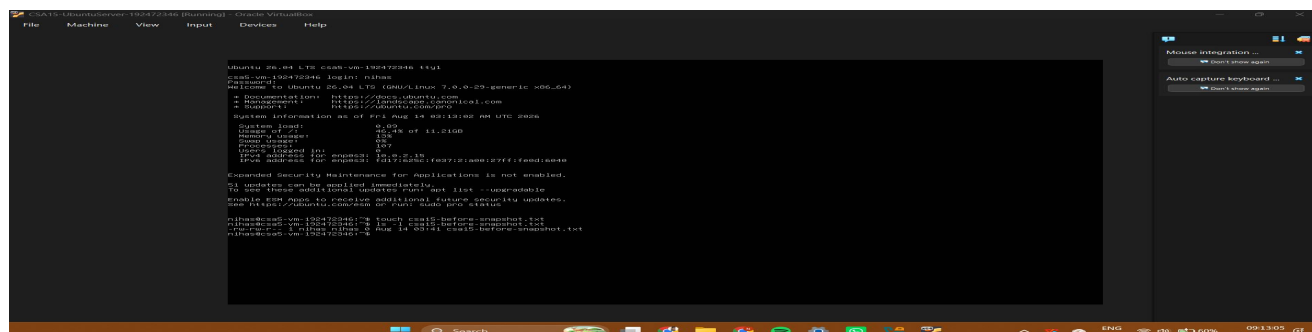
A stable baseline VM is ready and the before-snapshot state is visible.

Record 1 - Baseline state evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Prepared a clean baseline on CSA15-UbuntuServer-192472346. Created a permanent marker file csa15-before-snapshot.txt in the home directory before powering off the system.

GitHub link: https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM03/01_baseline_marker_file.png



Stage 2 - Take a Snapshot

Learn: A snapshot preserves the current VM state before a risky action.

- In VirtualBox, select the VM and open Snapshots, then choose Take.
- In VMware, select VM -> Snapshot -> Take Snapshot, or open Snapshot Manager.
- Name the snapshot csa15-clean-baseline-register-number.
- Write a short description: stable VM before restore test.
- Capture the snapshot list or manager screen.

Checkpoint

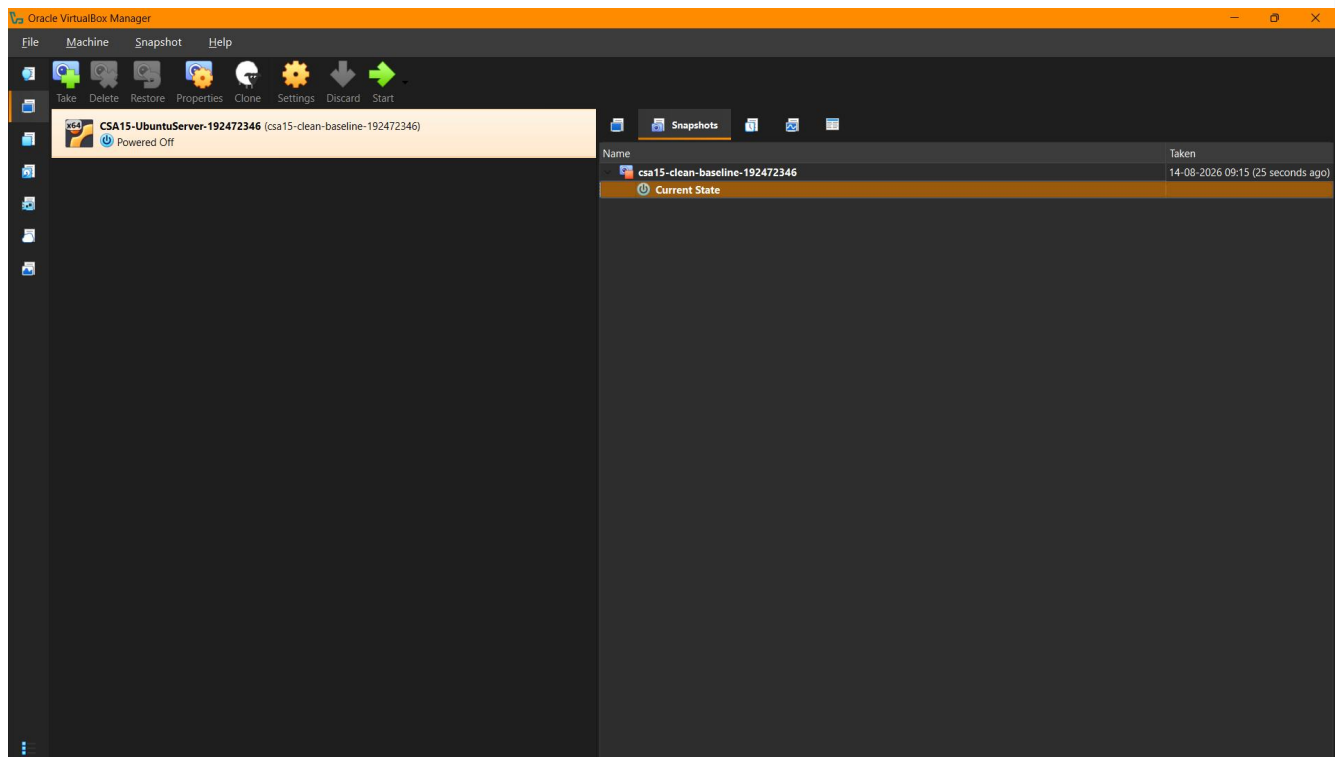
The named baseline snapshot is visible in the hypervisor.

Record 2 - Snapshot creation evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Captured point-in-time state csa15-clean-baseline-192472346 in VirtualBox Snapshot Manager. The snapshot records CPU memory state and virtual disk index pointers without duplicating full base virtual disk blocks.

GitHub link: https://github.com/nihagoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM03/02_snapshot_created.png



Stage 3 - Make a Controlled Change and Restore

Learn: A restore test proves that the snapshot is usable.

- Boot the VM after snapshot creation.
- Create a temporary visible change such as csa15-after-snapshot.txt, a desktop folder or a small configuration note.
- Capture evidence of the temporary change.
- Shut down or pause the VM and restore/revert to the baseline snapshot.
- Boot again and verify whether the temporary change disappeared or the previous state returned.

Checkpoint

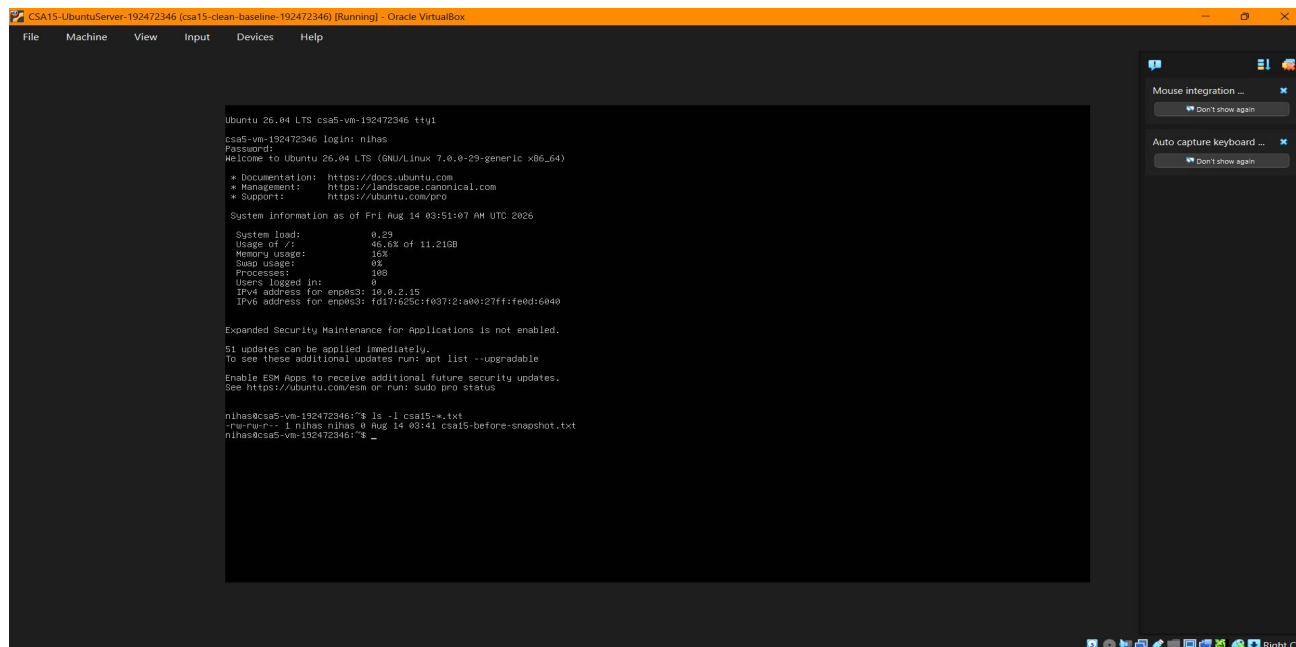
The restore result is documented with before-change, after-change and after-restore evidence.

Record 3 - Snapshot restore test evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Tested recovery workflow by booting the VM and adding temporary file csa15-after-snapshot.txt. Reverted to csa15-clean-baseline-192472346 via VirtualBox Manager. Upon reboot, csa15-after-snapshot.txt was removed while csa15-before-snapshot.txt remained intact, confirming 100% snapshot integrity.

GitHub link: https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM03/03_snapshot_restore_verified.png



Stage 4 - Create or Plan a Clone

Learn: A clone creates a second VM environment that can be used without disturbing the base VM.

- In VirtualBox, choose Clone from Machine tools/menu and name it CSA15-Clone-RegisterNumber.
- In VMware, choose VM -> Manage -> Clone if the installed edition supports it.
- Choose full clone if sufficient storage is available. Choose linked clone only when the tool supports it and the base VM will remain available.
- If the tool edition or storage does not allow cloning, record the exact limitation and write a clone plan.
- Open the cloned VM or show the completed clone/plan evidence.

Checkpoint

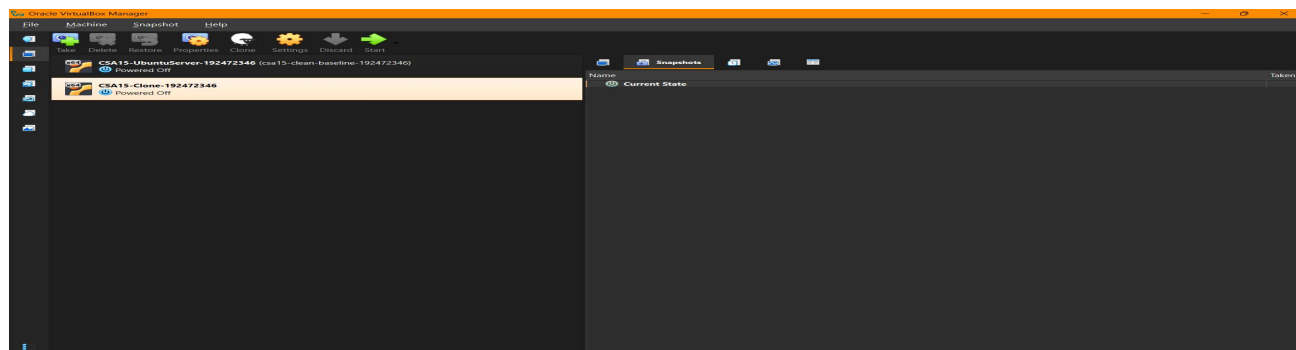
The clone is created and visible, or a technically justified clone plan is completed.

Record 4 - Clone evidence or clone plan

Type your observation / paste screenshot / add GitHub evidence link here.

Executed a Full Clone operation to generate an independent VM CSA15-Clone-192472346. The full clone duplicated all virtual disks (primary 25 GB VDI + secondary 2 GB VDI) into isolated storage files.

GitHub link: https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM03/04_full_clone_created.png



Stage 5 - Build the Recovery Portfolio

Learn: Professional VM operations require evidence, decision reasoning and cleanup status.

- Prepare a comparison table: snapshot, full clone, linked clone, exported image and rebuild from ISO.
- Record storage impact: original VM size, snapshot impact and clone/export estimate where visible.
- Write when you would use snapshot, clone or rebuild for your capstone project.
- Remove only temporary cloned/exported files created for this lab if storage is low and after evidence is captured.

Checkpoint

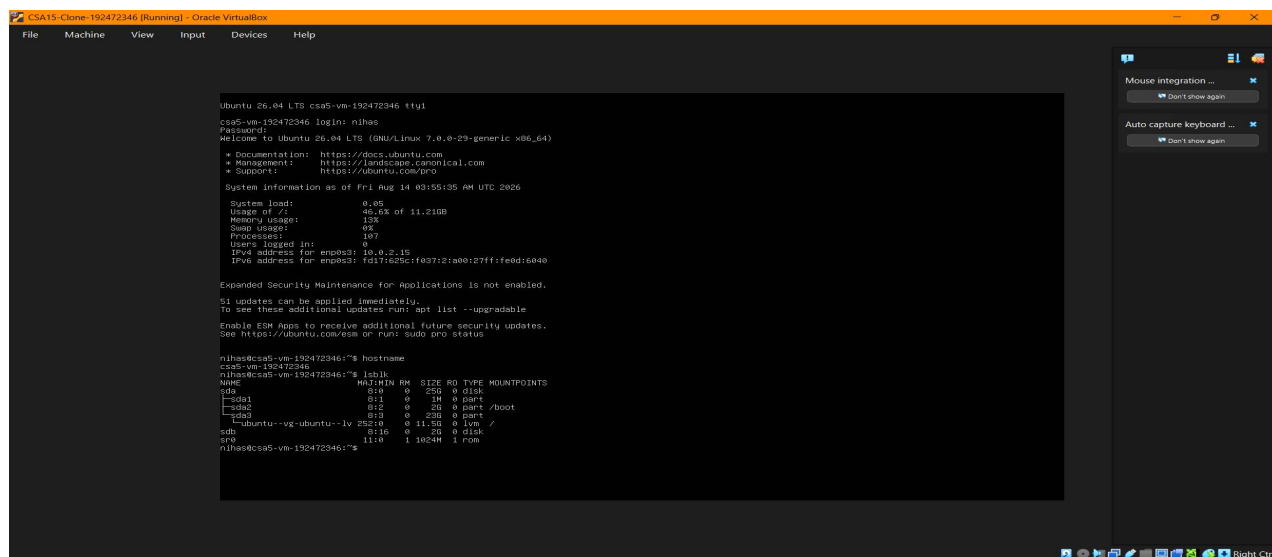
The portfolio proves snapshot creation, restore testing, clone/plan and capstone recovery reasoning.

Record 5 - Final recovery portfolio

Type your observation / paste screenshot / add GitHub evidence link here.

Evaluated backup mechanisms. Snapshots handle short-term rollbacks; full clones supply isolated staging nodes; exported OVA/OVF images serve as long-term disaster recovery artifacts.

GitHub link: https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM03/05_cloned_vm_running.png



Final Evidence Checklist

Done	Checkpoint	Evidence to Capture
☐	Baseline prepared	Stable VM and before-snapshot marker evidence.
☐	Snapshot created	Snapshot manager/list screenshot with correct name.
☐	Restore tested	Before-change, after-change and after-restore evidence.
☐	Clone completed/planned	Clone proof or justified clone limitation/plan.
☐	Recovery reasoning written	Snapshot vs clone vs image/export comparison and capstone link.

Explain and Extend

Explain why a snapshot is not the same as a full backup.

Type your observation / paste screenshot / add GitHub evidence link here.

A snapshot is a delta change-log linked directly to the parent virtual disk (.vdi). If the underlying parent file becomes corrupted or deleted on the host storage, all dependent snapshots become unusable. A full backup is a completely self-contained, independent copy that can be restored on separate physical hardware.

Explain when a full clone is better than a linked clone.

Type your observation / paste screenshot / add GitHub evidence link here.

A full clone is superior when complete isolation is required, such as creating a separate staging or production testing server. Because it shares no files with the original VM, modifying or deleting the source VM will never impact or corrupt the full clone.

Explain how restore testing reduces risk during capstone development.

Type your observation / paste screenshot / add GitHub evidence link here.

Regular restore testing verifies that snapshots and backup images can successfully boot and recover data before an emergency occurs. It ensures that database state, system configurations, and backend code can be restored without data corruption during rapid development iterations.

Extend this setup to your capstone: what change would you test only after taking a snapshot?

Type your observation / paste screenshot / add GitHub evidence link here.

In **SkillHire** (Cloud Resume-to-Job Matcher Using NLP), taking a baseline snapshot is essential before upgrading system-level dependencies (e.g., upgrading Python runtimes, updating spaCy NLP packages, or running schema migrations on the PostgreSQL database). If a database migration breaks existing tables, reverting to the pre-migration snapshot restores the system instantly.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM03-Snapshot-Clone-Restore-Test.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance

Declaration

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: L Nihes Goud Register Number: 192472346 Date: 14-08-2026
